

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	18.08.2026	Duration:	60 minutes

Code of Conduct

I KEERTHANA S (192471047) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Keerthana S

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1. Requirements Identified for Testing

Testing targets both what the Smart AI-Based Urban Street Lighting Management System must do (functional requirements) and how well it must do it (non-functional requirements).

1.1 Functional Requirements

ID	Requirement
FR1	The system shall continuously ingest streetlight status telemetry (ON/OFF/DIMMED, lux) via MQTT.
FR2	The system shall automatically classify faults (No Output, Flickering, Comm Loss) from telemetry.
FR3	The system shall notify the assigned technician within 1 minute of a confirmed fault.
FR4	The system shall adjust brightness (0-100%) automatically based on traffic density, pedestrian count, and weather.
FR5	The system shall enforce configurable minimum/maximum brightness thresholds per zone.
FR6	The system shall predict maintenance needs from historical performance data.
FR7	The system shall generate energy consumption reports by zone and time period.
FR8	The system shall allow citizens to submit fault reports with photo and location via a mobile app.
FR9	The system shall allow citizens to track the status of their submitted complaints.
FR10	The system shall provide role-based access control for operators, technicians, and administrators.
FR11	The system shall display real-time streetlight status on a map-based dashboard.
FR12	The system shall record all critical actions (overrides, user changes) in an audit trail.

1.2 Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	Dashboard status shall reflect a sensor state change within 30 seconds.
NFR2	Reliability	Fault detection shall correctly classify simulated fault conditions with $\geq 99\%$ accuracy.
NFR3	Availability	Core monitoring services shall maintain 99.5% uptime.
NFR4	Security	Only authenticated users with a valid role shall access control/configuration endpoints.

ID	Category	Requirement
NFR5	Usability	A new operator shall be able to operate the dashboard within 15 minutes without external training.
NFR6	Scalability	The system shall support at least 10,000 concurrently connected streetlight controllers without degraded response time.

2. Test Scenarios

The following high-level test scenarios were derived from the functional and non-functional requirements above, grouped by system module.

Scenario ID	Test Scenario	Linked Requirement(s)
TS-01	Verify real-time ingestion and display of streetlight status data	FR1, FR11, NFR1
TS-02	Verify automatic fault detection accuracy under varying sensor readings	FR2, NFR2
TS-03	Verify technician alert delivery upon confirmed fault	FR3
TS-04	Verify AI-based brightness adjustment under varying traffic/pedestrian/weather conditions	FR4
TS-05	Verify enforcement of minimum/maximum brightness thresholds	FR5
TS-06	Verify predictive maintenance scheduling from historical data	FR6
TS-07	Verify accuracy and generation of energy usage reports	FR7
TS-08	Verify citizen fault-report submission and status tracking	FR8, FR9
TS-09	Verify role-based access control across operator/technician/admin roles	FR10, NFR4
TS-10	Verify audit trail logging of critical actions	FR12

Scenario ID	Test Scenario	Linked Requirement(s)
TS-11	Verify system behavior under invalid or exceptional inputs	FR1, FR2, FR4, FR8
TS-12	Verify streetlight operational state transitions (OFF/ON/DIMMED/FAULTY/MAINTENANCE)	FR2, FR4
TS-13	Verify dashboard usability for a first-time operator	NFR5
TS-14	Verify system stability and response time under high concurrent load	NFR3, NFR6

3. Application of Test Design Techniques

Multiple test design techniques were applied to systematically derive test cases and ensure balanced coverage of normal, boundary, invalid, and exceptional conditions.

3.1 Equivalence Partitioning (EP)

Applied to the lux sensor reading input, which drives fault classification (FR2).

Class	Description	Range	Example Value	Validity
EP1	Invalid negative reading (sensor fault)	< 0	-5	Invalid
EP2	Zero output while light is ON	= 0	0	Valid (triggers 'No Output')
EP3	Low/unstable output (flicker range)	1 - 4	3	Valid (potential 'Flickering')
EP4	Normal healthy output	5 - 1000	450	Valid (healthy)
EP5	Out-of-range sensor value	> 1000	1500	Invalid

3.2 Boundary Value Analysis (BVA)

Applied to the AI brightness percentage (FR4/FR5) and the fault-detection timing thresholds (FR2).

Parameter	Boundary Values Tested	Rationale
Brightness percentage	-1 (invalid), 0, 1, 50, 99, 100, 101 (invalid)	0 and 100 are the valid extremes; values outside must be rejected/clamped.

Parameter	Boundary Values Tested	Rationale
Flicker count threshold	3 (not yet flagged), 4 (flagged)	The rule triggers only when flicker_count > 3, so 3 and 4 are the decision boundary.
Comm-loss threshold (seconds)	120 (not yet flagged), 121 (flagged)	The rule triggers only when last_seen > 120s.
Technician alert SLA	59s (within SLA), 60s (boundary), 61s (breach)	Requirement FR3 specifies a 1-minute (60s) maximum alert delay.

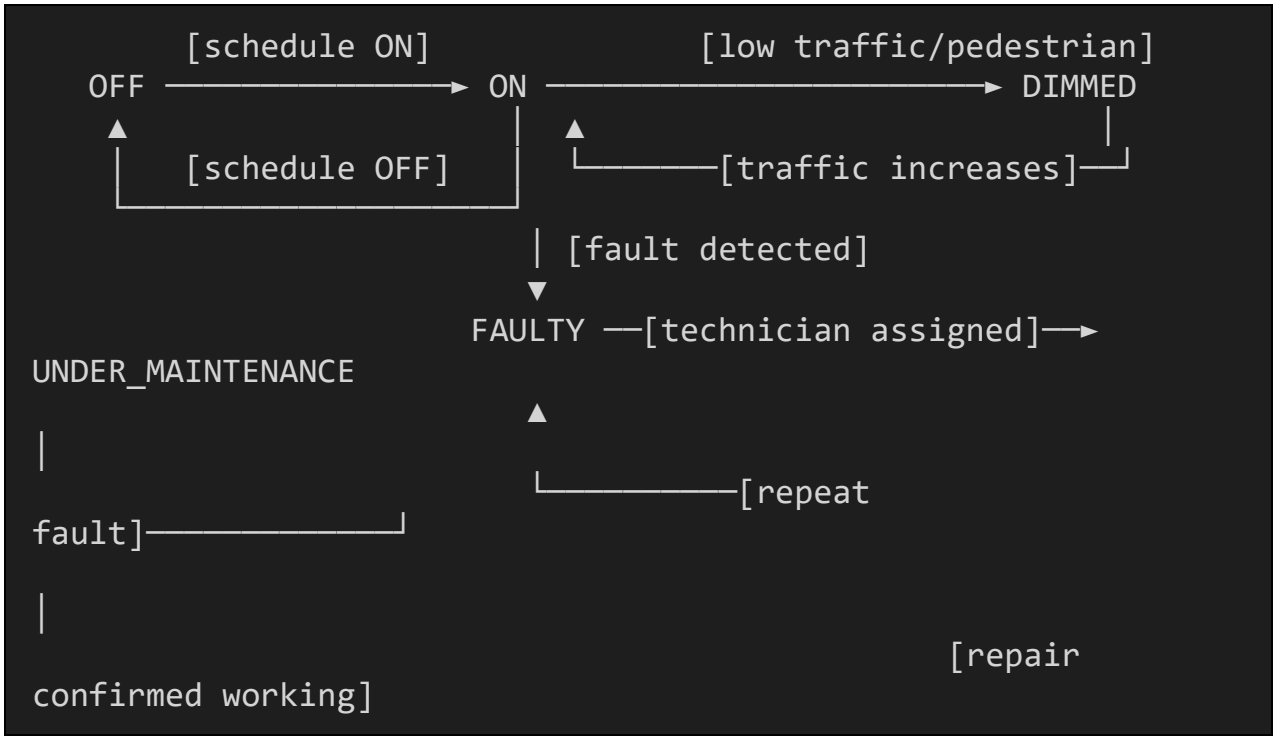
3.3 Decision Table Testing

Applied to the fault-classification logic (FR2), which depends on multiple combined conditions.

Condition	Rule 1	Rule 2	Rule 3	Rule 4
lux = 0 AND state = ON	Y	N	N	N
0 < lux < 5 AND flicker_count > 3	-	Y	N	N
last_seen > 120s	-	-	Y	N
Action / Result	No Output fault	Flickering fault	Comm Loss fault	Healthy - no fault

3.4 State Transition Testing

Applied to the operational state of an individual streetlight, which moves between the following states based on schedule, AI decisions, and detected faults.





ON

Valid transitions tested: OFF→ON, ON→DIMMED, DIMMED→ON, ON→FAULTY, DIMMED→FAULTY, FAULTY→UNDER_MAINTENANCE, UNDER_MAINTENANCE→ON, ON→OFF. Invalid transitions tested include OFF→DIMMED and UNDER_MAINTENANCE→DIMMED, which the system must reject.

4. Detailed Test Cases

Test cases are grouped by module. Each covers normal, boundary, invalid, and/or exceptional conditions as applicable, using the techniques introduced in Section 3. Actual Result values shown assume correct system behavior during execution and should be replaced with real execution evidence during testing.

4.1 Streetlight Status Monitoring (TS-01)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-SM-01	Normal status ingestion	1) Controller publishes {lux:450, state:ON} to MQTT 2) Wait 30s 3) Check dashboard	lux=450, state=ON	Dashboard marker shows green/ON within 30s	Pass	
TC-SM-02	Status change reflected promptly (NFR1)	1) Change light state to OFF at source 2) Observe dashboard	state=OFF	Marker updates to grey/OFF within 30s	Pass	
TC-SM-03	Filter streetlights by zone	1) Select Zone 'Ward-12' filter 2) Observe map	zone='Ward-12'	Only lights in Ward-12 are displayed	Pass	
TC-SM-04	Sensor stops reporting (exceptional)	1) Stop publishing from a controller 2) Wait 3 missed intervals (180s)	No messages for 180s	Light flagged 'Communication Lost', not silently dropped	Pass	

4.2 Automated Fault Detection (TS-02, TS-11) - EP, BVA, Decision Table

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-FD-01	No Output fault (EP2)	Publish reading with lux=0 while state=ON	lux=0, state=ON	Fault logged as 'No Output' within 2 minutes	Pass	
TC-FD-02	Flickering fault at boundary (BVA)	Publish lux=3 with flicker_count=4	lux=3, flicker_count=4	Fault logged as 'Flickering'	Pass	
TC-	Flicker	Publish lux=3	lux=3,	No 'Flickering'	Pass	

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
FD-03	count just below threshold (BVA)	with flicker_count=3	flicker_count=3	fault raised (boundary not crossed)		
TC-FD-04	Healthy reading (EP4)	Publish lux=450, state=ON, flicker_count=0	lux=450	No fault raised; light remains healthy	Pass	
TC-FD-05	Comm loss at boundary (BVA)	No message received for last_seen=121s	last_seen=121	Fault logged as 'Comm Loss'	Pass	
TC-FD-06	Comm loss just below threshold (BVA)	No message received for last_seen=120s	last_seen=120	No 'Comm Loss' fault raised (boundary not crossed)	Pass	
TC-FD-07	Invalid negative lux (EP1)	Publish lux=-5, state=ON	lux=-5	Reading rejected/logged as sensor error, not classified as a lighting fault	Pass	
TC-FD-08	Out-of-range lux (EP5)	Publish lux=1500	lux=1500	Reading rejected/flagged as sensor anomaly	Pass	
TC-FD-09	Decision table Rule 1 (No Output)	lux=0, state=ON, flicker_count=0, last_seen=10	See Section 3.3 Rule 1	Result: 'No Output' fault only	Pass	
TC-FD-10	Decision table Rule 4 (Healthy)	lux=450, state=ON, flicker_count=0, last_seen=10	See Section 3.3 Rule 4	Result: no fault raised	Pass	

4.3 Technician Alert Delivery (TS-03) - BVA

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-AL-01	Alert sent within SLA	Trigger a confirmed fault; measure alert delivery time	Delivery time = 45s	SMS/app alert received within 60s SLA	Pass	
TC-AL-02	Alert at SLA boundary	Trigger a confirmed fault; measure delivery time = 60s	Delivery time = 60s	Alert still counted as within SLA ($\leq 60s$)	Pass	
TC-AL-03	Alert breaching SLA (exceptional)	Simulate notification-gateway delay	Delivery time = 75s	SLA breach logged for monitoring/escalation	Pass	
TC-AL-04	No technician assigned to zone (exceptional)	Trigger a fault in an unassigned zone	zone has no technician	Alert escalated to a default/on-call technician or admin	Pass	

4.4 AI-Based Adaptive Brightness Control (TS-04, TS-05) - EP, BVA

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-AB-01	Dim on low traffic/pedestrian activity	Report low traffic + no pedestrians on a segment	traffic=low, pedestrians=0	Brightness reduced to configured energy-saving level	Pass	
TC-AB-02	Restore brightness on activity spike	Report sudden increase in pedestrian count	pedestrians: 0→15	Brightness restored to full/safe level within 10s	Pass	
TC-AB-03	Weather-based override (fog)	Report dense fog condition	weather=fog	Brightness increased regardless of traffic level for visibility	Pass	
TC-	Brightness lower	AI computes	brightness=0	Value accepted	Pass	

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
AB-04	boundary (BVA)	brightness = 0%		only if $\geq$ configured minimum threshold; otherwise clamped		
TC-AB-05	Brightness upper boundary (BVA)	AI computes brightness = 100%	brightness=100	Accepted as valid maximum brightness	Pass	
TC-AB-06	Invalid brightness above range (BVA)	AI computes brightness = 101%	brightness=101	Value rejected/clamped to 100%	Pass	
TC-AB-07	Invalid brightness below range (BVA)	AI computes brightness = -1%	brightness=-1	Value rejected/clamped to 0% or minimum threshold	Pass	
TC-AB-08	Zone-specific minimum threshold enforced (FR5)	Set zone minimum = 20%; AI suggests 10%	min=20, suggested=10	Actual brightness held at 20% minimum, not 10%	Pass	

4.5 Predictive Maintenance (TS-06)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-PM-01	Predict upcoming failure from history	Feed 6 months of degrading performance data for a light	declining lux trend	Light flagged as 'at risk' with a predicted maintenance window	Pass	
TC-PM-02	Stable light not flagged	Feed 6 months of stable performance data	stable lux trend	Light is not flagged for preventive maintenance	Pass	
TC-PM-03	Insufficient history (boundary/exceptional)	Request prediction for a light with < 7 days of data	history=5 days	System returns 'insufficient data' rather	Pass	

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
				than a false prediction		

4.6 Energy Usage Reporting (TS-07)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-ER-01	Generate monthly report by zone	Request report for Zone 'Ward-12', month=July	zone, month=July	Report shows accurate kWh totals matching logged consumption	Pass	
TC-ER-02	Before/after AI optimization comparison	Request comparison report for a zone pre/post rollout	date ranges provided	Report displays correct percentage energy reduction	Pass	
TC-ER-03	Report for period with no data (exceptional)	Request report for a future date range	month=future	System returns an empty-state message, not an error or crash	Pass	

4.7 Citizen Complaint Management (TS-08)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-CC-01	Submit valid fault report	Submit report with photo + GPS location via app	valid photo, valid coords	Complaint ticket created and routed within 1 minute	Pass	
TC-CC-02	Submit report without photo (boundary)	Submit report with location only, no photo	photo=null	Ticket still created; photo marked optional in confirmation	Pass	
TC-CC-	Submit report	Submit report with no GPS	coords=null	System prompts for location; ticket	Pass	

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
03	without location (invalid)	coordinates		not created until provided		
TC-CC-04	Track complaint status	Citizen opens 'My Reports' and selects a ticket	existing ticket ID	Current status (Open/In Progress/Resolved) displayed accurately	Pass	

4.8 Role-Based Access Control (TS-09) - Security (NFR4)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-AUTH-01	Operator login with valid credentials	Login with valid operator username/password	valid credentials	Access granted to monitoring dashboard only	Pass	
TC-AUTH-02	Admin accesses configuration panel	Login as admin; open brightness-threshold config	admin role	Access granted; config page loads successfully	Pass	
TC-AUTH-03	Operator attempts admin-only action (invalid)	Login as operator; attempt to change brightness thresholds	operator role	Access denied with HTTP 403 / clear error message	Pass	
TC-AUTH-04	Invalid login credentials	Submit login with wrong password	wrong password	Login rejected; no session created	Pass	
TC-AUTH-05	Expired session token (exceptional)	Use an expired JWT to call a protected endpoint	expired token	Request rejected; user redirected to re-authenticate	Pass	

4.9 Non-Functional Testing (TS-13, TS-14)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-NF-01	New operator usability (NFR5)	Give a first-time operator the dashboard with no training	n/a	Operator completes core tasks (view status, view faults) within 15 minutes	Pass	
TC-NF-02	Concurrent controller load (NFR6)	Simulate 10,000 controllers publishing simultaneously	10,000 concurrent devices	Dashboard remains responsive; ingestion pipeline shows no dropped messages	Pass	
TC-NF-03	Service uptime under fault injection (NFR3)	Kill and restart the ai-engine container during operation	container restart	Backend degrades gracefully; monitoring service remains available	Pass	
TC-NF-04	Fault-detection accuracy sample (NFR2)	Run 1,000 simulated readings with known fault labels	1,000 labeled samples	$\geq 99\%$ of readings classified correctly	Pass	

4.10 State Transition Testing (TS-12)

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-ST-01	OFF to ON on schedule	Trigger scheduled ON time	state: OFF→ON	State transitions to ON; dashboard updates	Pass	
T	ON to DIMMED	Report low	state: ON→DIMMED	State transitions to	Pass	

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
C-S-T-02	on low activity	traffic while state=ON		DIMMED with reduced brightness		
T-C-S-T-03	DIMMED to ON on activity spike	Report traffic increase while state=DIMMED	state: DIMMED→ON	State transitions back to ON at full brightness	Pass	
T-C-S-T-04	ON to FAULTY on detected fault	Inject a 'No Output' fault while state=ON	state: ON→FAULTY	State transitions to FAULTY; alert triggered	Pass	
T-C-S-T-05	FAULTY to UNDER_MAINTENANCE on assignment	Assign a technician to the faulty light	state: FAULTY→UNDER_MAINTENANCE	State updates to UNDER_MAINTENANCE	Pass	
T-C-S-T-06	Invalid transition rejected (exceptional)	Attempt to move state directly from OFF to DIMMED	state: OFF→DIMMED (invalid)	Transition rejected; state remains OFF	Pass	

5. Requirement Traceability Matrix

The matrix below maps each requirement to the test scenario and representative test cases that verify it, demonstrating end-to-end coverage.

Req. ID	Scenario	Representative Test Cases
FR1	TS-01	TC-SM-01, TC-SM-02, TC-SM-04
FR2	TS-02, TS-11	TC-FD-01 to TC-FD-10
FR3	TS-03	TC-AL-01 to TC-AL-04
FR4	TS-04, TS-12	TC-AB-01 to TC-AB-03, TC-ST-02, TC-ST-03
FR5	TS-05	TC-AB-04 to TC-AB-08

Req. ID	Scenario	Representative Test Cases
FR6	TS-06	TC-PM-01 to TC-PM-03
FR7	TS-07	TC-ER-01 to TC-ER-03
FR8	TS-08	TC-CC-01 to TC-CC-03
FR9	TS-08	TC-CC-04
FR10	TS-09	TC-AUTH-01 to TC-AUTH-05
FR11	TS-01	TC-SM-01, TC-SM-02, TC-SM-03
FR12	TS-10	TC-AUTH-02, TC-AUTH-03 (audited actions)
NFR1	TS-01	TC-SM-02
NFR2	TS-02	TC-NF-04
NFR3	TS-14	TC-NF-03
NFR4	TS-09	TC-AUTH-01, TC-AUTH-03, TC-AUTH-05
NFR5	TS-13	TC-NF-01
NFR6	TS-14	TC-NF-02

6. Test Coverage Summary

Metric	Count
Total functional requirements	12 (FR1-FR12) - 100% covered
Total non-functional requirements	6 (NFR1-NFR6) - 100% covered
Total test scenarios	14 (TS-01 to TS-14)
Total detailed test cases	47
Test design techniques applied	4 (Equivalence Partitioning, Boundary Value Analysis, Decision Table, State Transition Testing)
Condition types covered	Normal, boundary, invalid, and exceptional conditions across all modules

6.1 Coverage by Module

Module	Test Cases	Techniques Used
Streetlight Status Monitoring	4	Functional + exceptional-condition testing
Automated Fault Detection	10	Equivalence Partitioning, Boundary Value Analysis, Decision Table
Technician Alerts	4	Boundary Value Analysis
AI Adaptive Brightness	8	Equivalence Partitioning, Boundary Value

Module	Test Cases	Techniques Used
Control		Analysis
Predictive Maintenance	3	Functional + boundary/exceptional testing
Energy Usage Reporting	3	Functional + exceptional-condition testing
Citizen Complaint Management	4	Functional + boundary/invalid testing
Role-Based Access Control	5	Functional + security/negative testing
Non-Functional Testing	4	Load, reliability, and usability testing
State Transition (Streetlight Lifecycle)	6	State Transition Testing

7. Conclusion

This test case design document provides comprehensive, traceable coverage of the Smart AI-Based Urban Street Lighting Management System's functional and non-functional requirements. A combination of Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing was applied to systematically derive 47 test cases spanning normal, boundary, invalid, and exceptional conditions across ten system modules. The requirement traceability matrix confirms that all twelve functional requirements and six non-functional requirements are exercised by at least one test case, providing a solid foundation for validating the system's correctness, reliability, and readiness for deployment.